

Yegor Polyakov

AI Agent Platform Engineer · Backend / Full-Stack

Buenos Aires, Argentina · +54 11 2796-5334

egor-polyakov.vercel.app · egor.pol9kov@gmail.com

A backend engineer from fintech, then a full-stack product builder. I built Imperia OS solo — an AI platform whose agents write the code, while I hold the requirements and the spec map. My strength is the hunger for “why”, and the system carries the “how.”

Independent · Imperia OS · 2025 – present

Building toward an AI assistant that takes the shape its user needs, and the platform that makes it possible.

The agents that build the platform are the same agents its users get. A friend, a domain specialist, a developer, all assembled from domain modules. A module starts as business requirements that become a specification enforceably linked to the code, and a change that answers to no requirement cannot land. The system’s knowledge lives in that map, not in a developer’s head.

State machines govern an agent’s turn, a workflow, a deploy. Every step is explicit, either an LLM call or a deterministic operation, so a failure has a small blast radius, because the trace names the step and only what it touched replays.

An agent’s memory is history plus recall. As history grows, it folds into layered briefs, and the prompt stays within budget, with no layer repeating another. Recall surfaces what’s relevant by embedding match and places it in the context as hooks, not payloads, and the agent pulls the full thing when it matters.

The whole system reduces to three abstractions. Agents cover everything that acts. Blocks cover everything addressable, with one shape and one embedding. Protocols cover everything that crosses a boundary, and payment, signing and audit ride them as decorators.

Platform stack: TypeScript · Node.js · Next.js · Rust · PostgreSQL · MongoDB · Kuzu graph · Redis · Claude Agent SDK · MCP

Full-Stack Engineer · Independent · 2022 – 2025

A cross-platform mobile platform on Flutter and AWS. Clean Architecture from the first commit, cloud infrastructure as code in AWS CDK.

Dart and Flutter on the frontend, with feature-based modules, event-driven reactive layout and Riverpod for state. AWS Lambda in Go on the backend, with DynamoDB composite keys, S3 for media, Amplify and Cognito for auth.

Senior Software Engineer · IBM · 2020 – 2022

Built a full testing environment and CI/CD pipeline for Hyperledger Fabric, which doubled feature delivery speed. Refactored the project structure, cutting the codebase by 30%. Led the CouchDB→LevelDB migration that improved data-operation performance by 40%. Delivered a cross-channel encryption solution with zero defects, despite the absence of formal requirements.

Software Engineer · Insolar · 2019 – 2020

Designed the data architecture for a blockchain-based wallet, initiating the process and guiding a newly onboarded system architect through the key design decisions.

Software Engineer · Sber · 2018 – 2019

A custom CRUD-style abstraction layer over Hyperledger Fabric, self-initiated. New-feature implementation dropped from around eight hours to fifteen minutes, simplifying composite-key data operations.

St. Petersburg State University, Mat-Mekh · 2012 – 2017

Mathematics and Mechanics faculty. Arrived from LNMO, the Saint Petersburg laboratory for continuous mathematical education. Olympiad medals along the way in mathematics at ICYS and the City Olympiad, and in computer science at the Baltic Scientific and Engineering Competition. Worked through university in parallel on Java, Android, Delphi, and embedded C++ for spectrograph instrumentation at a materials research institute.